Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements

Step 1: List Creation

1(a) Create age_list

```
[1]
✓ 0s
age_list = [24, 25, 26, 27, 28]

print("Age List:", age_list)

Age List: [24, 25, 26, 27, 28]
```

1(b) Create name_list

```
name_list = ["Arun", "Priya", "Kavi", "Divya", "Rahul"]

print("Name List:", name_list)

Name List: ['Arun', 'Priya', 'Kavi', 'Divya', 'Rahul']
```

```
age_list = [24, 25, 26, 27, 28]

print("Age List:", age_list)

name_list = ["Arun", "Priya", "Kavi", "Divya", "Rahul"]

print("Name List:", name_list)

Age List: [24, 25, 26, 27, 28]
Name List: ['Arun', 'Priya', 'Kavi', 'Divya', 'Rahul']
```

Step 2: List Operations / Modifications

2(a) Append "Yazhini" to name_list

```
name_list.append("Yazhini")

print("After Append:", name_list)

After Append: ['Arun', 'Priya', 'Kavi', 'Divya', 'Rahul', 'Yazhini']
```

2(b) Insert 30 at index 2 in age_list

```
age_list.insert(2, 30)

print("After Insert:", age_list)
```

```
After Insert: [24, 25, 30, 26, 27, 28]
```

2(c) Remove "Yazhini" from name_list

```
name_list.remove("Yazhini")

print("After Remove:", name_list)
```

```
After Remove: ['Arun', 'Priya', 'Kavi', 'Divya', 'Rahul']
```

2(d) Pop the last element from age_list

```
age_list.pop()

print("After Pop:", age_list)
```

```
After Pop: [24, 25, 30, 26, 27]
```

2(e) Extend age_list with [29, 30, 26]

```
age_list.extend([29, 30, 26])

print("After Extend:", age_list)
```

```
After Extend: [30, 27, 26, 25, 24, 29, 30, 26]
```

2(f) Sort age_list in descending order

```
age_list.sort(reverse=True)

print("Descending Order:", age_list)
```

```
Descending Order: [30, 30, 29, 27, 26, 26, 25, 24]
```

2(g) Find Maximum, Minimum and Sum

```
print("Maximum Age:", max(age_list))
print("Minimum Age:", min(age_list))
print("Sum of Ages:", sum(age_list))
```

```
Maximum Age: 30
Minimum Age: 24
Sum of Ages: 217
```

Step 3: Accessing List Elements

3(a) Print the first element of name_list

```
name_list = ["Arun", "Priya", "Kavi", "Divya", "Rahul"]
print("First Element:", name_list[0])
```

```
First Element: Arun
```

3(b) Print the last element of name_list

```
print("Last Element:", name_list[-1])
```

```
Last Element: Rahul
```

3(c) Print elements from index 2 to index 4

```
print("Elements from index 2 to 4:", name_list[2:5])
```

```
Elements from index 2 to 4: ['Kavi', 'Divya', 'Rahul']
```

3(d) Print elements in reverse order

```
print("Reverse Order:", name_list[::-1])
```

```
Reverse Order: ['Rahul', 'Divya', 'Kavi', 'Priya', 'Arun']
```

Step 4: Dictionary

4(a) Create a dictionary named student_marks

```
student_marks = {
    "Arun": 75,
    "Priya": 90,
    "Kavi": 85,
    "Divya": 78,
    "Rahul": 88
}

print("Student Marks:", student_marks)
```

```
Student Marks: {'Arun': 75, 'Priya': 90, 'Kavi': 85, 'Divya': 78, 'Rahul': 88}
```

4(b) Access and print a specific student's mark

```
print("Priya's Mark:", student_marks["Priya"])
```

```
Priya's Mark: 90
```

4(c) Add a new student "Janani" with 80 marks

```
student_marks["Janani"] = 80

print("After Adding Janani:", student_marks)
```

```
After Adding Janani: {'Arun': 75, 'Priya': 90, 'Kavi': 85, 'Divya': 78, 'Rahul': 88, 'Janani': 80}
```

4(d) Update an older student's mark to 82

```
student_marks["Arun"] = 82

print("After Updating Arun's Mark:", student_marks)
```

```
After Updating Arun's Mark: {'Arun': 82, 'Priya': 90, 'Kavi': 85, 'Divya': 78, 'Rahul': 88, 'Janani': 80}
```

4(e) Use keys(), values() and items()

```
print("Student Names:", student_marks.keys())
print("Student Marks:", student_marks.values())
print("Student Details:", student_marks.items())
```

```
Student Names: dict_keys(['Arun', 'Priya', 'Kavi', 'Divya', 'Rahul', 'Janani'])
Student Marks: dict_values([82, 90, 85, 78, 88, 80])
Student Details: dict_items([('Arun', 82), ('Priya', 90), ('Kavi', 85), ('Divya', 78), ('Rahul', 88), ('Janani', 80)])
```

Step 5: Set Operations

5(a) Create a set named my_set

```
my_set = {'a', 'e', 'i', 'o', 'u', 'a', 'a', 'i'}

print("My Set:", my_set)
```

```
My Set: {'o', 'e', 'u', 'a', 'i'}
```

5(b) Attempt to change the value using an index

```
my_set.add('s')
print("After Adding s:", my_set)
```

```
After Adding s: {'s', 'o', 'e', 'u', 'a', 'i'}
```

5(c) Create two sets

```
set1 = {1, 3, 5, 7, 9}

set2 = {2, 3, 5, 8, 10}

print("Set 1:", set1)
print("Set 2:", set2)
```

```
Set 1: {1, 3, 5, 7, 9}
Set 2: {2, 3, 5, 8, 10}
```

5(d) Calculate Union and Intersection

```
print("Union:", set1.union(set2))


print("Intersection:", set1.intersection(set2))
```

```
Union: {1, 2, 3, 5, 7, 8, 9, 10}
Intersection: {3, 5}
```

Step 6: Operators & Conditional Statements (IF, ELIF, ELSE)

```
score = int(input("Enter your score (0 to 10): "))


if score < 0 or score > 10:
    print("Invalid Score! Please enter a score between 0 and 10.")


elif score > 7:
    print("Above Average: Excellent performance! Keep it up!")

elif score >= 4:
    print("Average: Good effort! Keep practicing, there's room for improvement.")

else:
    print("Below Average: Need to improve your performance. Consistent practice will lead to better results.")
```

```
Enter your score (0 to 10): 5
Average: Good effort! Keep practicing, there's room for improvement.
```